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M.Tech. DEGREE EXAMINATIONS, MAY 2026

Second Semester

21EIE602T - EMBEDDED CONTROL SYSTEMS

(For the candidates admitted from the academic year 2024 - 2025 & 2025 - 2026 onwards)

Time: Three Hours

Max. Marks: 100

(5 x 20 = 100 Marks)

Marks BL CO

Answer **ANY FIVE** Questions

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| 1. a) Explain a System on Chip (SoC) and its architecture. (15 Marks) | 20 | 2 | 1 |
| b) Differentiate between microcontroller and SoC. (5 Marks) | | | |
| 2. (a) Describe the role of bus structures in embedded systems architecture. (10 Marks) | 20 | 2 | 1 |
| (b) Explain how different types of buses facilitate communication between the processor, memory, and peripheral devices. (5 Marks) | | | |
| (c) Explain the TCP/IP protocol stack in the context of embedded systems and IoT applications and discuss how embedded devices utilize the TCP/IP stack for reliable communication in networked environments. (5 Marks) | | | |
| 3. a) Recognise an embedded system and explain its key characteristics with examples. (10 Marks) | 20 | 2 | 1 |
| b) Identify between general-purpose computing systems and embedded systems. (5 Marks) | | | |
| c) Explain the architecture of a typical embedded system with a neat block diagram. (5 Marks) | | | |
| 4. a) Interpret Emulation and explain how it differs from pure software simulation. (10 marks) | 20 | 3 | 2 |
| b) Implement a Hardware-in-the-Loop (HIL) setup where the ECU "thinks" it is connected to a real engine. Summarize the primary implementation challenges of HIL? (10 marks) | | | |
| 5. (a) Demonstrate how Heuristic Evaluation can be applied to assess the usability of the interface using appropriate usability heuristics. | 20 | 3 | 2 |
| (b) Identify and explain the key interface requirements that must be considered when designing user interfaces for embedded or industrial systems. | | | |
| (c) Discuss how the results of heuristic evaluation can be used to improve the interface design and overall user experience. | | | |
| 6. Consider a smart home automation system that controls lighting, temperature, and security functions. | 20 | 3 | 3 |

- (a) Interpret the system using appropriate UML diagrams such as Use Case Diagram, Class Diagram, and Sequence Diagram.
- (b) Explain how each UML diagram represents different aspects of the system including system functionality, structural relationships, and interaction between components.
- (c) Discuss how UML modeling helps in analyzing, designing, and documenting complex embedded or software systems.

7. (a) Sketch and explain the primary building blocks of SDL, namely System, Block, Process, and Signal. 20 3 3
- (b) Illustrate how these components interact with each other to model the behavior and communication within a complex embedded system.
